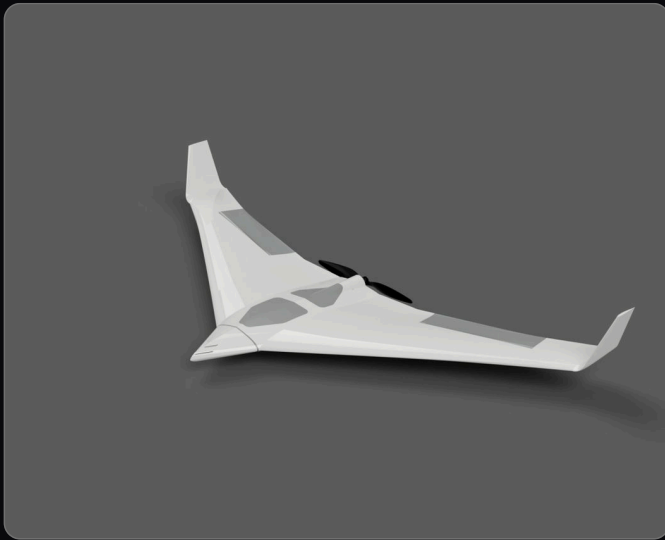


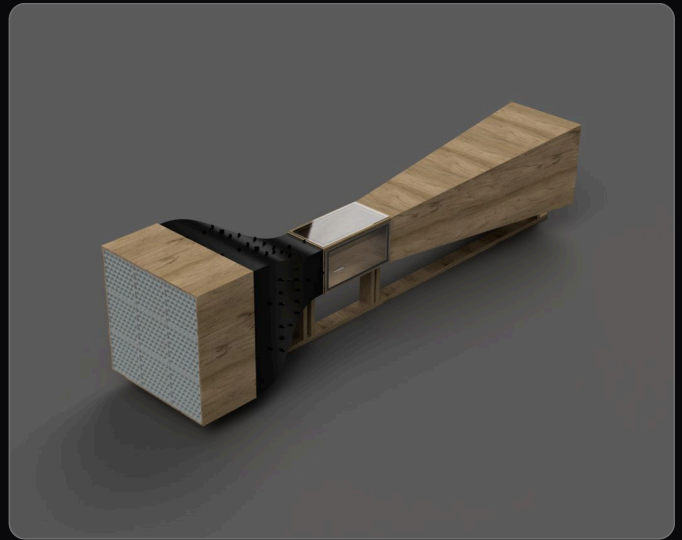
# Xueji (Jackie) Zou

Mechanical Engineering Student focused on Aerodynamics

I model in CFD, build the hardware, and test it. My current projects are a flying-wing UAV and a low-speed wind tunnel I built to check the simulation.



01 Flying-wing UAV



02 Low-speed open-circuit wind tunnel

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EDUCATION

University of Kansas  
B.S. Mechanical Engineering, expected  
May 2029

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EDUCATION

### University of Kansas

B.S. Mechanical Engineering · Expected May 2029 · GPA 4.0 / 4.0

- Dean's List, 2025 to 2026
- American Society of Mechanical Engineers (ASME), 2025 to 2026
- Chem-E-Car, Fall 2025

CONTENTS

#### 01 Flying-wing UAV

A 42 in flying wing built for long endurance, with autonomous flight planned next.

#### 02 Low-speed open-circuit wind tunnel

A wind tunnel I designed and built to measure the flying wing's lift and drag and compare them with CFD.

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SKILLS

#### CAD and analysis

SolidWorks, Fusion 360,  
AutoCAD, XFLR5, ANSYS Fluent

#### Programming and controls

MATLAB, C++, Arduino IDE,  
ESP32, INAV

#### Additional

3D printing, Microsoft Excel,  
English and Mandarin

# Flying-wing UAV

## A tailless wing. A harder stability problem.

A 42 in blended flying wing built for long endurance and printed in modules, so damaged parts can be replaced quickly.

### SCOPE

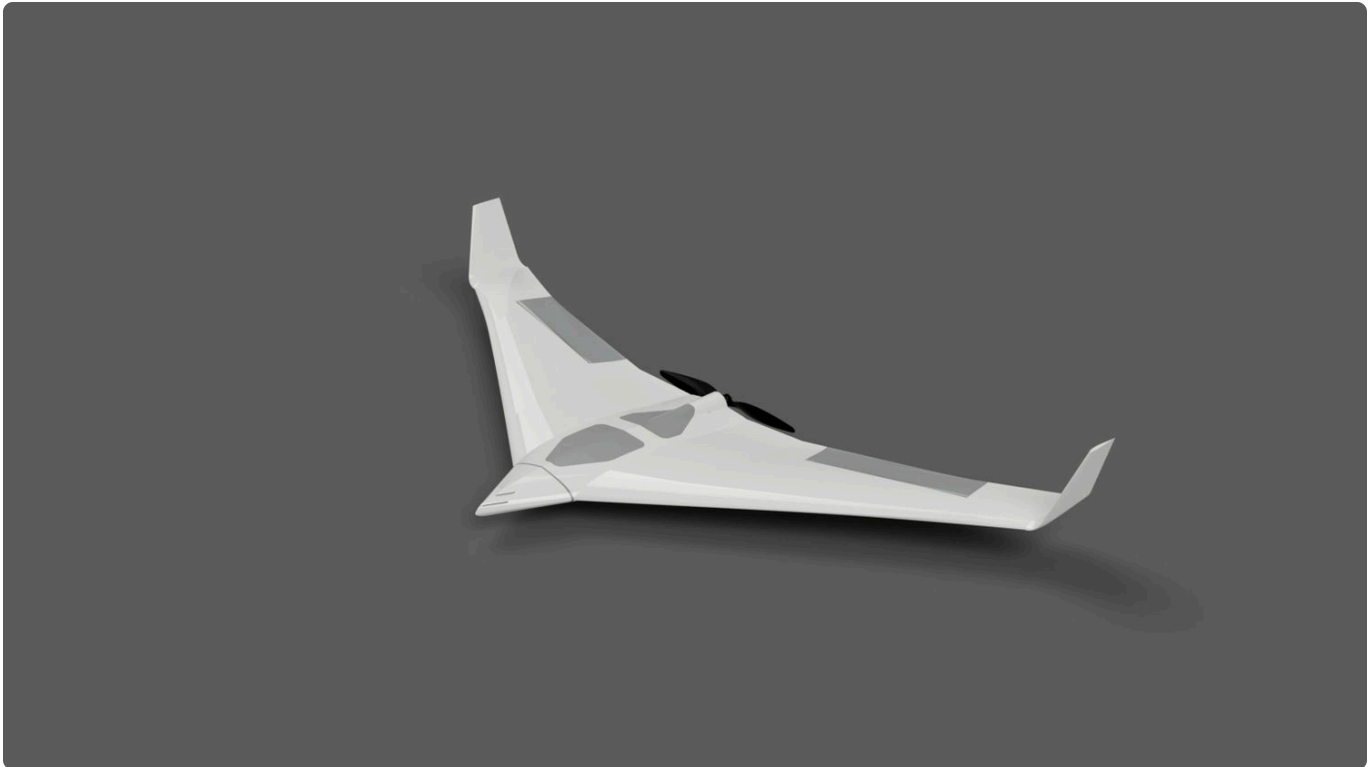
Airfoil and planform design, CFD, a 3D-printed airframe, propulsion, and avionics integration.

### STATUS

In progress. Flight testing continues, and the current weight is 850 g.

### EVIDENCE

XFLR5 and ANSYS Fluent results, CAD, and built-airframe test records. The 37.87% drag reduction is a CFD result, not a flight measurement.



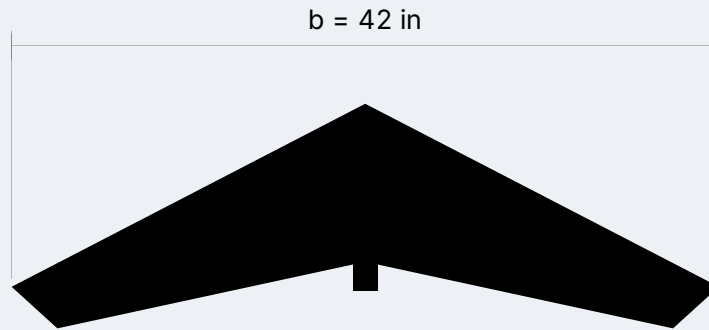
Flying-wing UAV, CAD render.

### 01 THE CHALLENGE

## More lifting surface, higher efficiency.

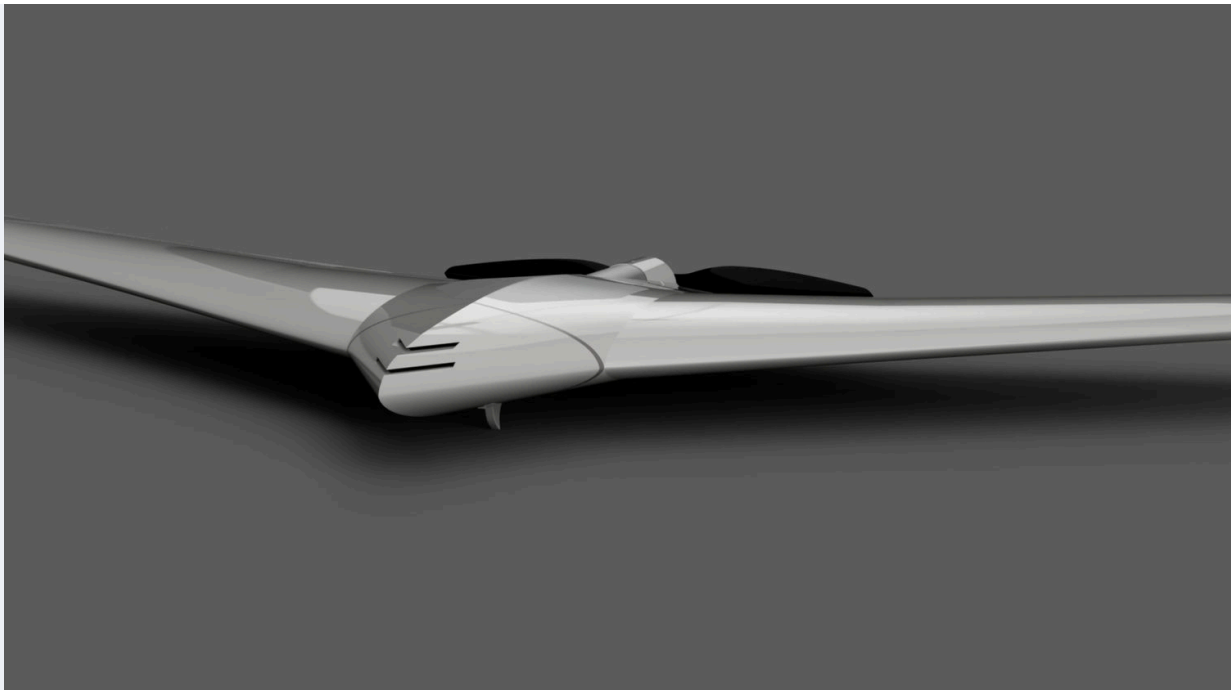
A typical aircraft is a tube with wings. The body generates no lift and only adds parasitic drag.

A blended wing design maximizes its lifting surface. But without a tail to carry the vertical and horizontal stabilizers, pitch and yaw stability become significantly harder to achieve.



#### Planform schematic

A swept flying wing with no fuselage tube and no tail. Schematic, 42 in span.



#### WING-BODY BLEND

The blended wing root and pylon, with no separate fuselage tube.

## 02 OBJECTIVES

### **Modular and repairable, with maximum flight time.**

The wing was sized to carry 1.5 kg. That holds all of its components and leaves room for future upgrades such as cameras and other sensors, while balancing size against weight.

Crashes are inevitable, so the plane is built in modules. Broken parts can be printed and replaced quickly, which keeps the time between iterations short.

## DESIGN PAYLOAD

## WINGSPAN

PRINTED AIRFRAME MASS



## PANEL BONDING

### 03 PLANNING

## PITCH

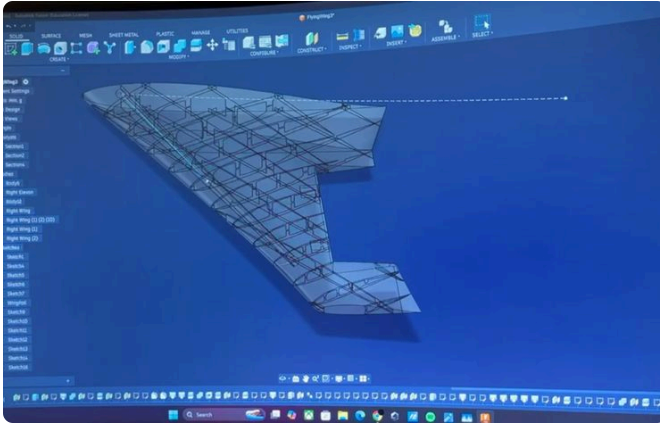
ROLL

YAW

## DESIGN NOTES

## STABILITY ANALYSIS

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#### CAD MODEL

The wing modeled in Fusion 360, with internal ribs for the printed structure.

### 04 INTEGRATION

## Printing, assembling, testing, and repeat.

The entire plane is 3D printed, so any part that breaks in testing can be reprinted and swapped in.

#### Airframe

Printed on an Elegoo Centauri Carbon in vase mode with a specialized lightweight filament. The 42 in wingspan airframe came in under 250 g.

#### Propulsion

One Hobbywing Skywalker 980KV brushless motor, a 3200 mAh LiPo battery, and a 10 × 5.5 in propeller chosen to maximize loiter time.

#### Avionics

A Matek F405-WING running INAV, with a nano receiver, servos for the control surfaces, and a BZ-251 GPS.

#### Propeller testing

Three propeller types were tested on a custom static thrust bench that measured thrust, current, voltage, and power to compare their efficiency.



#### STATIC THRUST TEST

Static thrust testing setup for the motor and propeller, logging thrust, voltage, current, and power.



#### POST-CRASH LAYOUT

The airframe and electronics laid out after a crash, checked before the next iteration.

### 05 PROBLEMS AND SOLUTIONS

# Each solution comes with its own problem.

Three problems from the build, each with the fix and the new problem that fix created.

## PROBLEM 1

### Body too large

#### PROBLEM

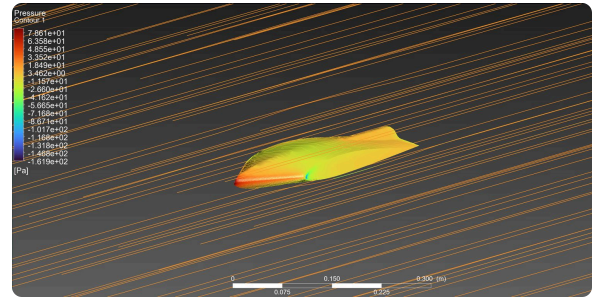
The body was too large. It created unnecessary empty space and was not as streamlined or as blended into the wing as it could have been.

#### SOLUTION

Built a slimmer, thinner body that blends into the wing. CFD in ANSYS Fluent showed the drag coefficient dropped by 37.87%.

#### NEW PROBLEM

The slimmer body has weak points and is more prone to damage in a crash.



#### CFD PRESSURE CONTOUR

ANSYS Fluent pressure contour and streamlines used to confirm the 37.87% drag reduction.

## PROBLEM 2

### Difficult launches

#### PROBLEM

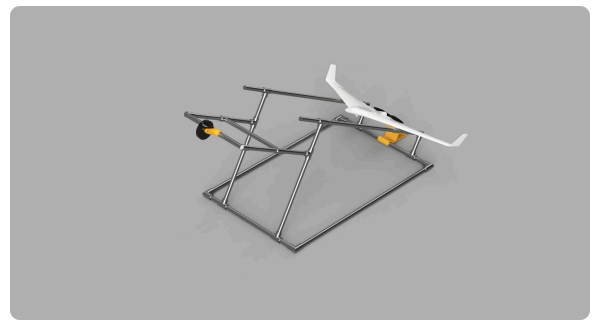
Because of its size, the plane was hard to launch safely and consistently. That caused many crashes that could have been prevented.

#### SOLUTION

Built an RC launcher from PVC pipe and surgical tubing that pulls the plane with 4 kg of force.

#### NEW PROBLEM

The servos hit the bump from the tee connections and stripped teeth off the gears. The launch surface was also too low, so the plane had to clear it before the motor throttled up.



#### LAUNCHER CAD

The PVC and surgical-tubing launcher, modeled before it was built.

#### PROBLEM 3

### Fragile wing

#### PROBLEM

The carbon-fiber spars did not reach the ends of the wing, so the tips were fragile and broke off after nearly every crash.

#### SOLUTION

Instead of buying new rods, redesigned the wing 1.5 in shorter at the ends so the existing rods reach closer to the tips.

#### NEW PROBLEM

The shorter wing increased wing loading.



#### WING TIP DAMAGE

Broken tip sections after a crash, the kind of damage that led to shortening the wing.

#### 06 STATUS

## Progress is still ongoing.

The airframe and the full CAD assemblies keep changing as more trials are run.

Next, the launcher will be changed so the motor can stay on before launch, and components will be changed to cut up to 150 g from the current weight.

**850 g**

CURRENT WEIGHT

**150 g**

PLANNED REDUCTION, AT MOST

## BILL OF MATERIALS

| ITEM                           | QTY | WEIGHT  | PRICE    |
|--------------------------------|-----|---------|----------|
| AIRFRAME                       |     |         |          |
| Printed shell                  | 1   | < 250 g | —        |
| Carbon Fiber Rods              | 2   | 35 g    | \$22.99  |
| Mounting hardware              | —   | —       | —        |
| PROPULSION                     |     |         |          |
| Hobbywing Skywalker 2316 980KV | 1   | 79 g    | \$29.99  |
| Hobbywing Skywalker V2 50A     | 1   | 37 g    | \$19.99  |
| APC 1050                       | 1   | 15 g    | \$3.99   |
| Liperior 3300mAh 3S LiPo       | 1   | 60 g    | \$23.99  |
| AVIONICS                       |     |         |          |
| Matek F405-WING V2             | 1   | 25 g    | \$89.99  |
| BZ-251                         | 1   | 11 g    | \$20.99  |
| Radiomaster RP3 V2 2.4GHz      | 1   | 4.6 g   | \$19.99  |
| Radiomaster Boxer ELR          | 1   | —       | \$153.99 |
| E-flite 9g Plastic Servos      | 2   | 18 g    | \$6.99   |
| TEST AND LAUNCH EQUIPMENT      |     |         |          |
| Static thrust bench            | 1   | —       | —        |
| Launcher                       | 1   | —       | —        |

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## TECHNICAL RECORD

|                  |                                                   |
|------------------|---------------------------------------------------|
| CONFIGURATION    | Blended flying wing, no tail                      |
| PRINTED AIRFRAME | Under 250 g                                       |
| DESIGN PAYLOAD   | 1.5 kg                                            |
| DRAW REDUCTION   | 37.87% in CFD (ANSYS Fluent)                      |
| WASHOUT ANGLE    | 3°                                                |
| WING SWEEP       | 33°                                               |
| AVIONICS         | Matek F405-WING (INAV), nano receiver, BZ-251 GPS |
| STATUS           | In progress                                       |

|                  |                                                                        |
|------------------|------------------------------------------------------------------------|
| WINGSPAN         | 42 in                                                                  |
| CURRENT WEIGHT   | 850 g                                                                  |
| ENDURANCE TARGET | 45+ minutes                                                            |
| STABILITY TOOLS  | XFLR5, then Fusion 360 and ANSYS Fluent                                |
| DIHEDRAL ANGLE   | 2°                                                                     |
| PROPULSION       | One Hobbywing Skywalker 980KV, 10 × 5.5 in propeller, 3S 3200 mAh LiPo |
| LAUNCHER         | PVC and surgical tubing, 4 kg pull                                     |

# Wind tunnel

## Measure it. Then check the simulation.

A low-speed, open-circuit wind tunnel constructed from commercially available materials to measure the lift and drag of the flying-wing UAV and validate its CFD predictions.

### SCOPE

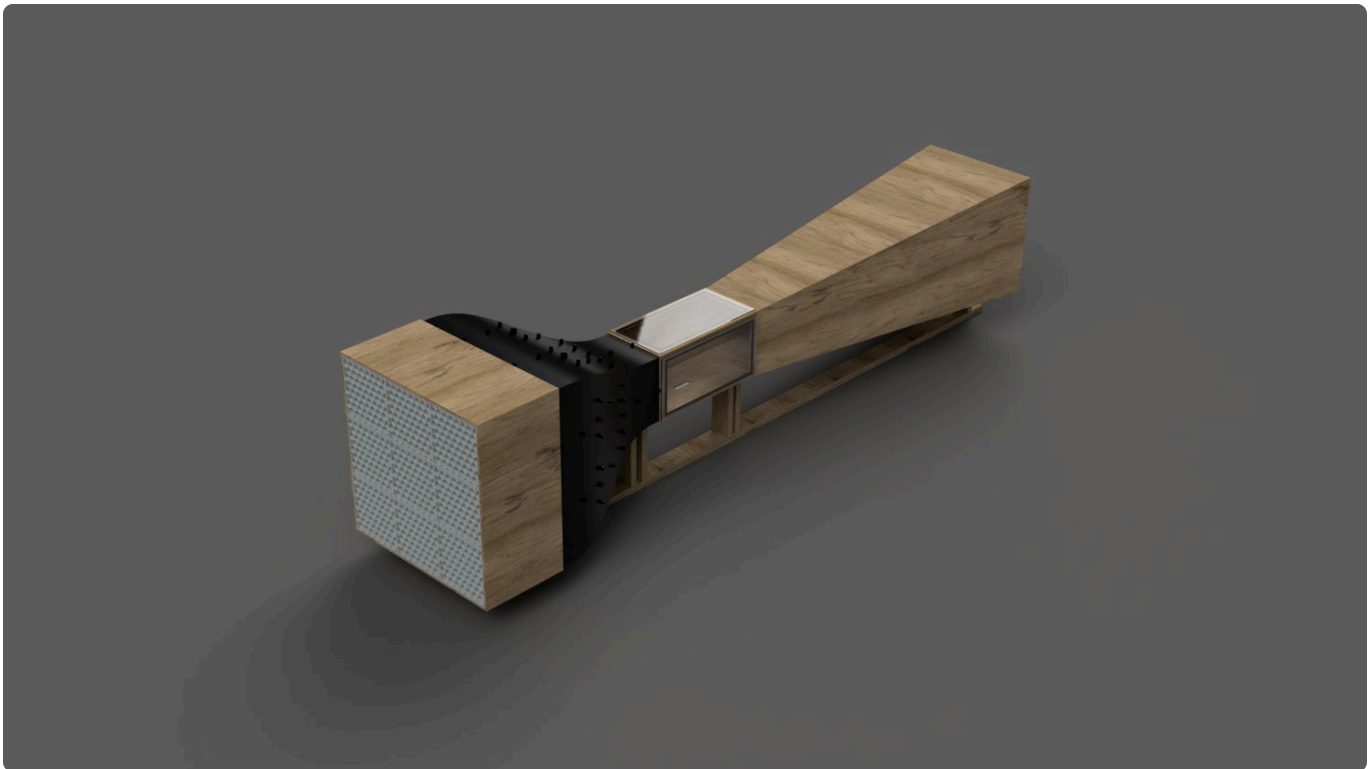
Tunnel sizing and construction, flow conditioning, contraction design and fabrication, airspeed measurement, a two-axis force balance, and data reduction.

### STATUS

In progress. Blockage correction and Reynolds-number matching remain outstanding.

### EVIDENCE

Sensor data are logged through an Arduino DAQ and processed in MATLAB. The comparison with ANSYS CFD has not been done yet.



Low-speed wind tunnel, CAD render.

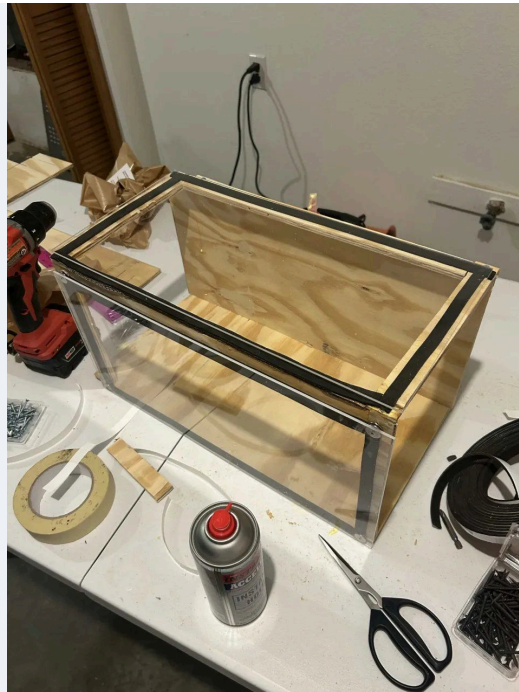
### 01 THE CHALLENGE

## Too big, or too small?

The test section must be large enough to accommodate the intended test models.

However, a larger test section requires a longer diffuser, increasing manufacturing difficulty. It also reduces the achievable airspeed for a given fan and substantially increases the material required for the contraction cone and honeycomb.

The final dimensions therefore balance these constraints, allowing the tunnel to replicate the CFD conditions while being constructed entirely from commercially available materials.



#### TEST SECTION

The 9 × 9 × 18 in test section during assembly, sealed with weatherstrip tape.

## 02 OBJECTIVES

### A simulation is only a prediction.

CFD relies on turbulence models to approximate the Navier-Stokes equations, so its results require experimental validation.

Across the design iterations of the flying wing's center body, CFD predicted a reduction in drag coefficient from 0.4212 to 0.2617, a 37.87% decrease. The wind tunnel was developed to verify results of this kind.

Design requirements: construction from commercially available materials; a test-section velocity of 15 m/s (33 mph) to match the CFD conditions; accurate measurement of both lift and drag; and a model frontal area no greater than 10% of the test-section cross-sectional area.

**0.4212 →  
0.2617**

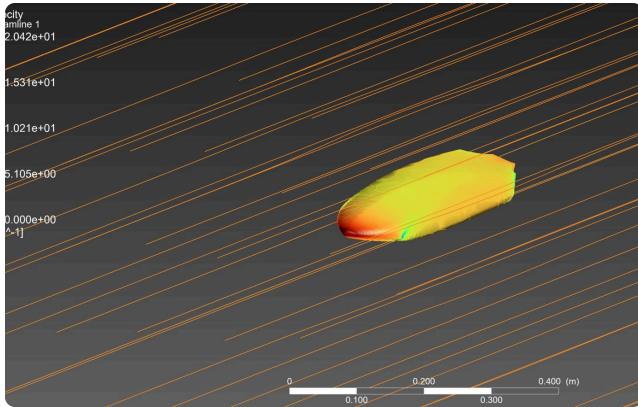
CFD DRAG COEFFICIENT,  
FIRST TO FINAL

**15 m/s**

TARGET AIRSPEED (33  
MPH)

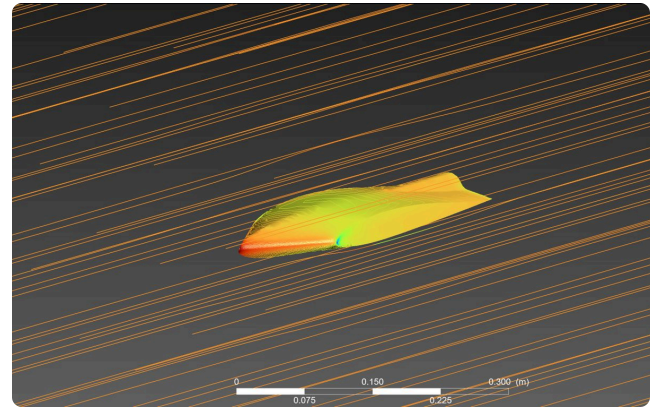
**≤ 10%**

MAXIMUM BLOCKAGE RATIO



#### OLD MODEL

The original center body in ANSYS Fluent, velocity streamlines. CFD drag coefficient 0.4212.



#### CURRENT MODEL

The current, slimmer center body in ANSYS Fluent, pressure contour. CFD drag coefficient 0.2617.

### 03 PLANNING

## Size it once, then build around it.

Using NASA's Wandering Wind Tunnel as a reference, a  $9 \times 9 \times 18$  in test section was selected, followed by a 55 in diffuser with a  $3^\circ$  half angle.

The contraction cone profile follows Bell and Mehta's 5th-order polynomial. It was initially designed with a 9:1 contraction ratio ( $27 \times 27$  in inlet,  $9 \times 9$  in outlet); the inlet was later reduced to  $24 \times 24$  in, as described in Problems and Solutions.

A high-CFM fan was required to reach the target velocity, and a two-axis load cell arrangement was required to measure lift and drag independently.

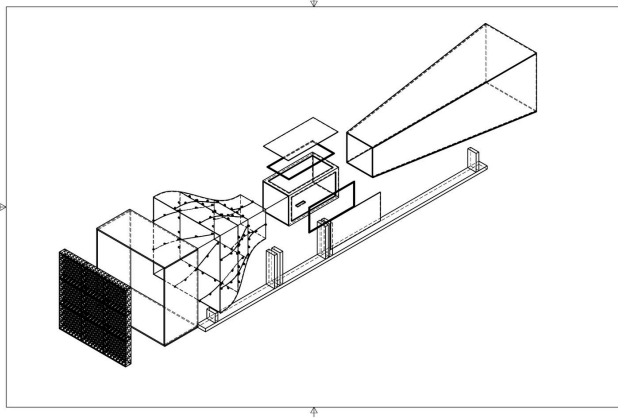
**STRAIGHTEN** Printed honeycomb

**CONDITION** Two fiberglass mesh screens

**ACCELERATE** 5th-order polynomial contraction (Bell and Mehta)

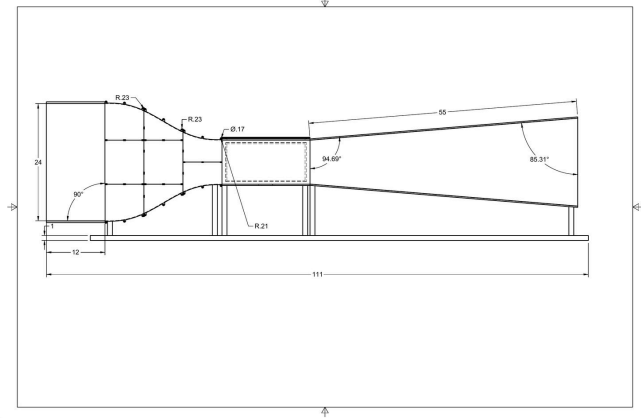
**TEST**  $9 \times 9 \times 18$  in test section

**RECOVER** 55 in diffuser,  $3^\circ$  half angle



#### EXPLODED ASSEMBLY

Exploded view of every section, from the honeycomb to the diffuser.



#### SIDE VIEW DRAWING

Dimensioned side view in inches: 111 in overall, 24 in inlet, 55 in diffuser.

## 04 INTEGRATION

### Cut, print, seal, and wire.

All wooden components, including the diffuser, test section, settling chamber, and support legs, were cut with a circular saw. Weatherstrip tape was applied at every joint to keep the tunnel airtight.

Acrylic panels on the top and side of the test section are secured with neodymium magnets, allowing quick removal for model access.

#### Contraction cone

3D printed in 20 segments and assembled with M4 fasteners.

#### Flow conditioning

A honeycomb printed in 9 segments (0.75 in cells, 2.5 in deep) straightens the flow. Two fiberglass mesh screens spaced 4 in apart downstream of the honeycomb dissipate small-scale eddies.

#### Fan

Variable-speed portable fan rated at 2894 CFM.

#### Electronics

An Arduino was selected over an ESP32 because all sensors operate at a 5 V output. An MPXV7002DP differential pressure sensor connected to a pitot-static tube provides airspeed.

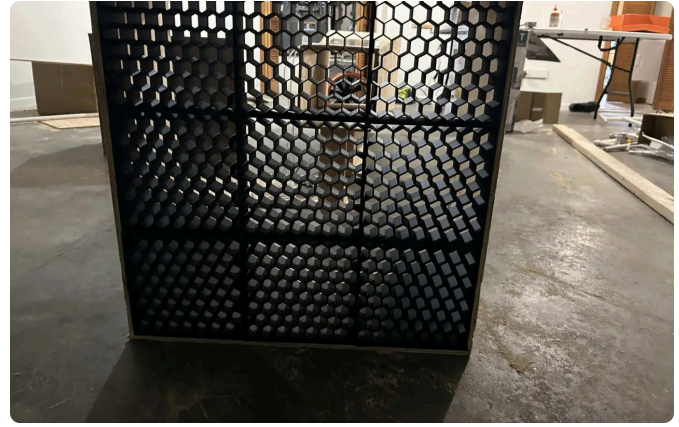
#### Force balance

1 kg load cells on 3D-printed mounts measure lift and drag along two axes.



#### SECTIONS BUILT

The settling chamber, test section, and diffuser cut and assembled.



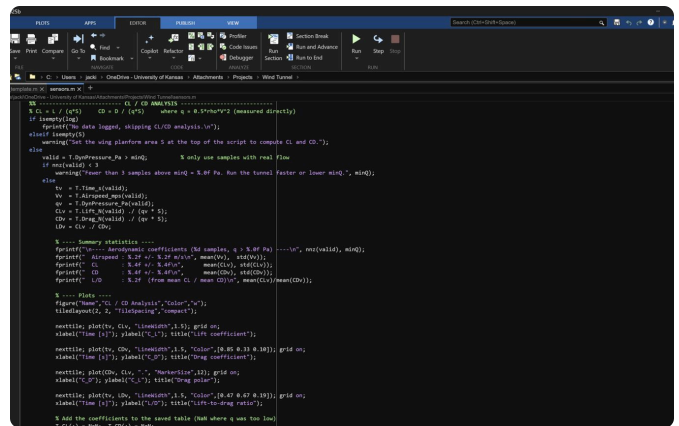
#### HONEYCOMB

The honeycomb, printed in 9 segments and framed in plywood.



#### 2-AXIS LOAD CELL

The 2-axis load cell setup that measures lift and drag.



#### MATLAB DAQ

The MATLAB DAQ script that reads the Arduino sensors and computes CL, CD, and L/D.

## 05 PROBLEMS AND SOLUTIONS

### Each solution comes with its own problem.

Each issue encountered during construction is documented with its solution and any resulting trade-off.

## PROBLEM 1

### PLA glass transition

#### PROBLEM

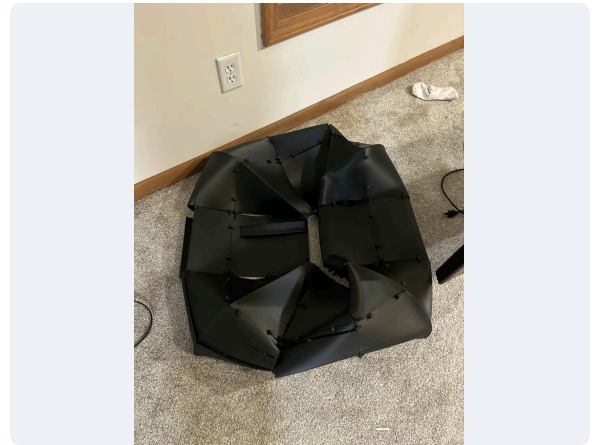
During transport to campus, the 9:1 PLA contraction cone deformed in the bed of a truck after exceeding PLA's glass transition temperature range of 131 to 149 °F.

#### SOLUTION

A smaller contraction cone with a 7:1:1 ratio (24 × 24 in inlet, 9 × 9 in outlet) was printed to reduce print time and material usage.

#### NEW PROBLEM

The lower contraction ratio provides less flow acceleration between the inlet and the test section.



#### DEFORMED CONE

The 9:1 cone after it softened in the truck.

## 06 STATUS

### Ongoing.

#### BILL OF MATERIALS

| ITEM                                               | QTY     | PRICE            |
|----------------------------------------------------|---------|------------------|
| STRUCTURE                                          |         |                  |
| Plywood, 4 × 8 ft, 11/32 in thick                  | 4       | \$22.88 each     |
| Acrylic glass (test section top and side)          | 2       | \$11.59 each     |
| Neodymium magnets                                  | 4       | \$8.99           |
| Weatherstrip tape                                  | —       | —                |
| FLOW CONDITIONING                                  |         |                  |
| Contraction cone (3D printed PLA, 20 parts, 7:1:1) | 8 rolls | \$10.99 per roll |
| M4 nuts and screws                                 | —       | —                |
| Honeycomb (3D printed PLA, 9 parts)                | 4 rolls | \$10.99 per roll |
| Fiberglass window mesh                             | 2       | —                |
| AIRFLOW                                            |         |                  |
| VEVOR portable fan, variable speed, 2894 CFM       | 1       | \$149.99         |
| INSTRUMENTATION                                    |         |                  |
| MPXV7002DP differential pressure sensor            | 1       | \$29.99          |
| Pitot-static tube                                  | 1       | \$13.99          |
| 1 kg load cells with HX711 amplifiers              | 2       | \$15.99          |

## TECHNICAL RECORD

|                   |                                                                                         |                     |                                                  |
|-------------------|-----------------------------------------------------------------------------------------|---------------------|--------------------------------------------------|
| TYPE              | Low-speed, open-circuit                                                                 | REFERENCE DESIGN    | NASA Wandering Wind Tunnel                       |
| TEST SECTION      | 9 × 9 × 18 in                                                                           | TARGET AIRSPEED     | 15 m/s (33 mph)                                  |
| AIRSPEED MEASURED | Up to 32 mph                                                                            | BLOCKAGE LIMIT      | Model area ≤ 10% of test section area            |
| CONTRACTION       | Bell and Mehta 5th-order polynomial, 7:1:1 (24 × 24 in to 9 × 9 in), first built at 9:1 | DIFFUSER            | 55 in, 3° half angle                             |
| FLOW CONDITIONING | Honeycomb (0.75 in cells, 2.5 in deep) and two fiberglass screens 4 in apart            | FAN                 | Variable-speed portable fan, 2894 CFM            |
| FORCE BALANCE     | 2-axis, 1 kg load cells with HX711 amplifiers                                           | SOFTWARE            | Arduino (C++) for logging, MATLAB for processing |
| CAD               | SolidWorks                                                                              | COMPARISON WITH CFD | Not done yet                                     |
| STATUS            | In progress                                                                             |                     |                                                  |